NSO8062 Geofüüsikaline andmetöötlus

Loeng 6: Andmete jaotus. Tuuleandmed.

Stella-Theresa Luik

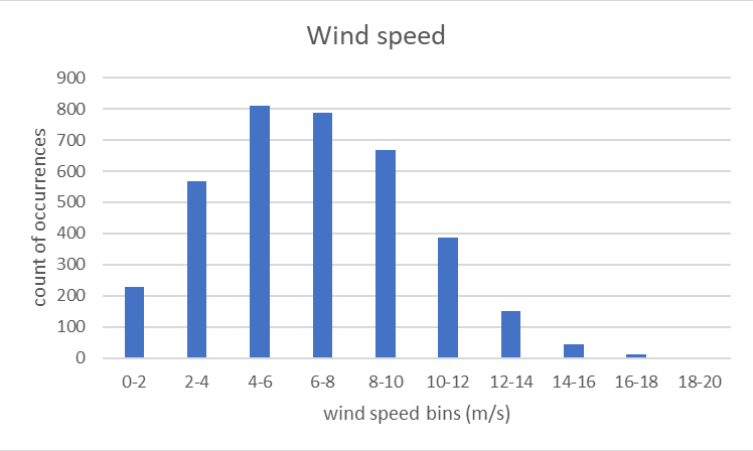
12.03.2026

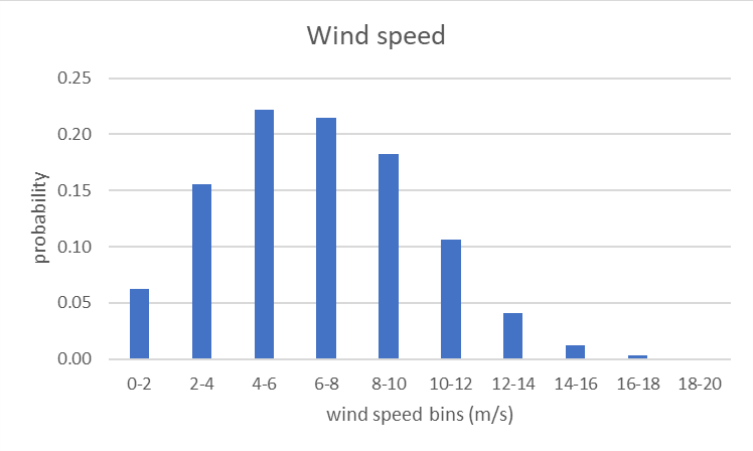
Ajakava ja teemad

Nädal	Teema	Õppejõud
1	Sissejuhatus ainesse	S-T. Luik
2	Erinevad andmete tüübid ja lihtsamad tegevused nendega.	S-T. Luik
3	NetCDF failide loomine, muutmine jne.	S-T. Luik
4	Keri andmed. Profiilide visualiseerimine, andmete eemaldamine, silumine, gradientide leidmine.	S-T. Luik
5	Andmete 1D interpoleerimine ja korrelatsioonid, statistikud jne.	S-T. Luik
6	Andmete jaotus. Tuuleandmed ja nende visualiseerimine	S-T. Luik
7	2D andmed, interpoleerimine, visualiseerimine (nt kaart)	
8	Kuumalained	
9	TBD	
10	TBD	
11	TBD	
12	TBD	
13	TBD	
14	TBD	
15	Konsultatsioon	
16	**Eksam**	
17		
18		
19	10/06/2026 kevadsemestri eksamite sooritamise lõpptähtaeg	
20	15/06/2026 kevadsemestri lõpp	

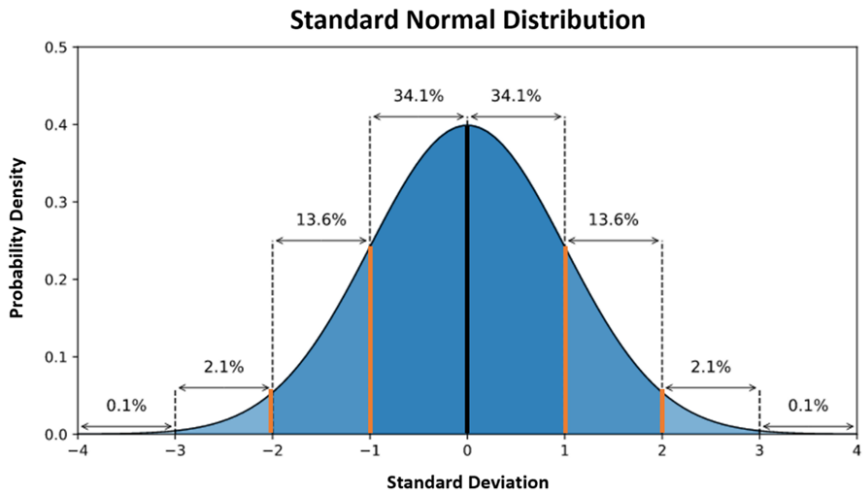
Why do we need to know the distribution of the data?

- In statistics, the distribution tells you **where the values fall** and **how often they occur.**
- Knowing your data's distribution is like knowing the "personality" of your dataset. If you assume your data is calm and predictable (Normal) when it's actually wild and erratic (Skewed), your statistical conclusions will be flat-out wrong.
- Most classic statistical tools operate on the assumption that your data follows a Normal Distribution (the Bell Curve).
- If you use statistical tools on non-normal data, e.g., the mean could become a villain: In skewed data, the "average" is often pulled away from the reality of the majority. *But then again, if you have a lot of data then the mean would describe the reality pretty good.*





Common data distributions

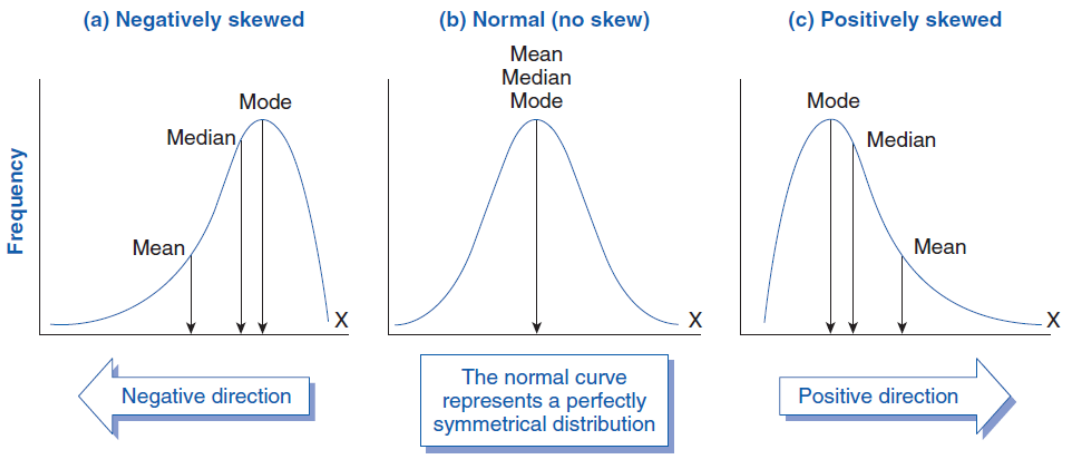


1. Normal distribution, e.g. SST (sea surface temperature)

In a stable environment, daily SST over a long period of time tend to follow a Normal Distribution.

- The scenario: If you measure the temperature of the Gulf Stream in July over 50 years, most readings will cluster around a mean (e.g., 28°C).
- Why it matters: Scientists use this to define "anomalies." If a temperature reading falls 3 standard deviations away from the mean, it’s a statistical signal of a potential El Niño event or a marine heatwave.
- The shape: A classic, symmetrical bell curve.

Common data distributions

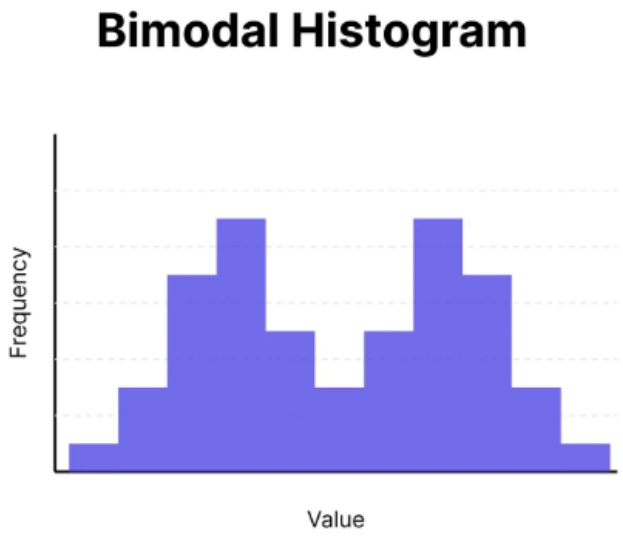


2. Skewed distribution, e.g. phytoplankton biomass

Biological data in the ocean is almost always Right-Skewed (Positive Skew).

- The scenario: For most of the year, phytoplankton levels (measured by Chlorophyll-a) are relatively low. However, during a "Spring Bloom," the population explodes.
- Why it matters: If you use a simple "average" to calculate carbon sequestration, the bloom periods will make the ocean look much more productive than it is on a typical day. Oceanographers often use a log-transformation to turn this "wild" data into a normal shape for easier analysis.
- The shape: A tall peak at low values with a long "tail" stretching toward the high-value blooms.

Common data distributions



3. Bimodal Distribution, e.g., mixed layer depth (MLD)

The "Mixed Layer" is the top layer of the ocean where wind and waves mix the water to a uniform temperature. This often shows a Bimodal Distribution when looking at an entire year.

- The scenario:
  - Peak 1: In Summer, the sun heats the surface, creating a very shallow mixed layer.
  - Peak 2: In Winter, storms and cooling create a very deep mixed layer.
- Why it matters: If you took the average of these two peaks, you'd get a "middle" depth that actually never occurs in nature. You must treat these as two distinct seasonal states.
- The shape: Two distinct humps, one for the summer state and one for the winter state.

Weibull distribution and wind data

- Continuous probability distribution, that can fit an extensive range of distribution shapes.
- If you simply say "my wind data is skewed," you are just describing how it looks. If you say "my wind data follows a Weibull distribution," you are making a functional claim that allows you to do advanced engineering, such as:
  - Extrapolation: Predicting the "50-year extreme wind" by extending the mathematical tail.
  - Parameterization: Summarizing 10 years of daily data into just two numbers (k and c) for a map.
- Your wind speed data is **naturally skewed** because it is physically impossible to have negative wind (a "wall" at zero) but possible to have very high speeds (the "tail").
- The **Weibull distribution** is the specific mathematical "glove" that fits that "hand" of skewed data perfectly.

https://doi.org/10.24381/cds.adbb2d47

Wind data from ERA5

ERA5 is the fifth generation ECMWF reanalysis for the global climate and weather for the past 8 decades. Data is available from 1940 onwards.

Reanalysis combines model data with observations from across the world into a globally complete and consistent dataset using the laws of physics.

ERA5 provides hourly estimates for a large number of atmospheric, ocean-wave and land-surface quantities.

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PROGRAMME OF THE EUROPEAN UNION · Copernicus · Climate Change Service · IMPLEMENTED BY ECMWF

European Centre for Medium-Range Weather Forecasts

Climate Data Store · Datasets · User guide · Applications · Forum · Live status

Datasets > ERA5 hourly data on single levels from 1940 to present

ERA5 hourly data on single levels from 1940 to present

Overview · Download · Documentation

Data description

Data type	Gridded
Projection	Regular latitude-longitude grid
Horizontal coverage	Global
Horizontal resolution	Reanalysis: 0.25° x 0.25° (atmosphere), 0.5° x 0.5° (ocean waves) Mean, spread and members: 0.5° x 0.5° (atmosphere), 1° x 1° (ocean waves)
Temporal coverage	1940 to present
Temporal resolution	Hourly
File format	GRIB
Update frequency	Daily

Wind data from ERA5

fileinfo | fileinfo.Variables

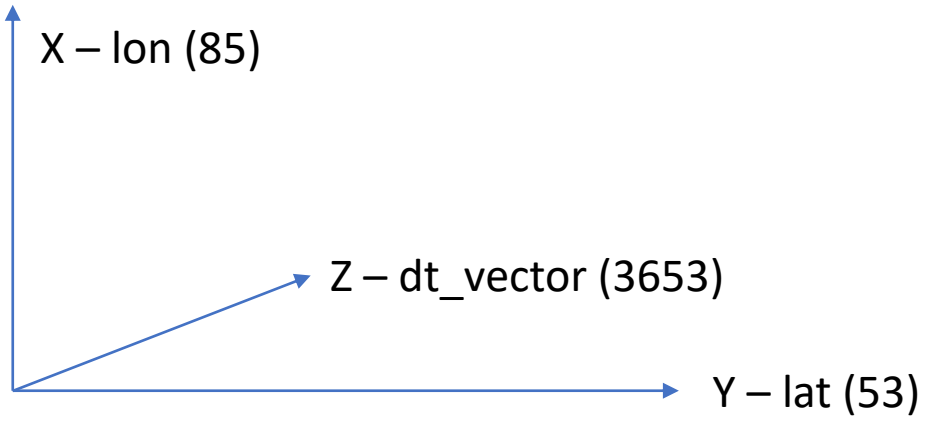
fileinfo.Variables

Fields	Name	Dimensions	Size	Datatype	Attributes	ChunkSize	FillValue	DeflateLevel	Shuffle
1	'number'	[]	1	'int64'	1x3 struct	[]	-92233720...	[]	0
2	'valid_time'	1x1 struct	3653	'int64'	1x4 struct	[]	-92233720...	[]	0
3	'latitude'	1x1 struct	53	'double'	1x5 struct	[]	NaN	[]	0
4	'longitude'	1x1 struct	85	'double'	1x4 struct	[]	NaN	[]	0
5	'expver'	1x1 struct	3653	'string'	[]	[]	""	[]	0
6	'u10'	1x3 struct	[85,53,3653]	'single'	1x34 struct	[43,27,1827]	NaN	1	1
7	'v10'	1x3 struct	[85,53,3653]	'single'	1x34 struct	[43,27,1827]	NaN	1	1
8	'sst'	1x3 struct	[85,53,3653]	'single'	1x33 struct	[43,27,1827]	NaN	1	1
9	'lsm'	1x3 struct	[85,53,3653]	'single'	1x34 struct	[43,27,1827]	NaN	1	1
10									

fileinfo | fileinfo.Variables | fileinfo.Variables(2).Attributes

fileinfo.Variables(2).Attributes

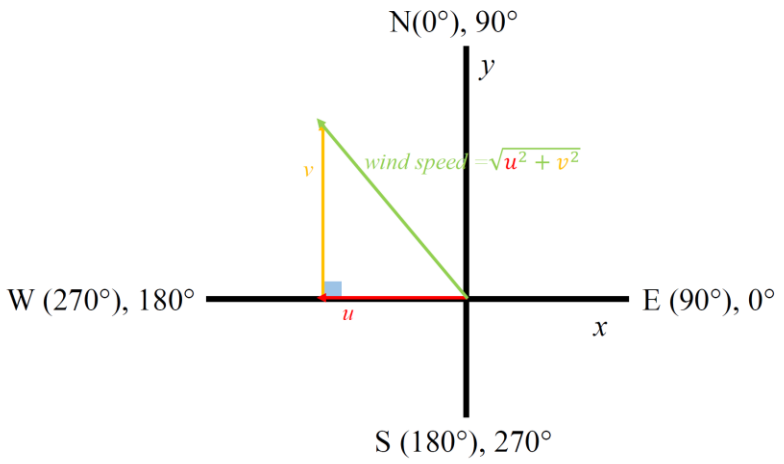
Fields	Name	Value
1	'long_name'	'time'
2	'standard_name'	'time'
3	'units'	'seconds since 1970-01-01'
4	'calendar'	'proleptic_gregorian'



Workspace

Name	Value
dataFolder	'C:\Users\Stella\OneDrive - Ta
dt_vector	3653x1 datetime
fileinfo	1x1 struct
fileName	'data_stream-oper_stepType-i
fullPath	'C:\Users\Stella\OneDrive - Ta
lat	53x1 double
lon	85x1 double
lsm	85x53x3653 double
sst	85x53x3653 double
time	3653x1 int64
u10	85x53x3653 double
v10	85x53x3653 double

Wind speed and direction



- Wind data vectors:
  - u10 - This parameter is the eastward component of the 10m wind.
  - v10 - This parameter is the northward component of the 10m wind.

- Wind speed
  - WS = sqrt($u10^2$+$v10^2$);

- Wind direction (mathematical – where the wind is blowing to, North = 0 degrees, 360 degrees).
  - WD = mod(atan2d(u10, v10), 360);

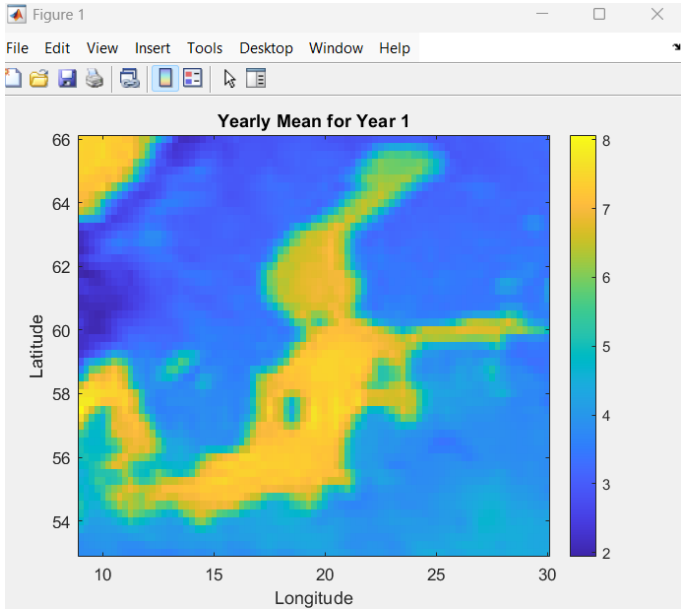
Lecture 6. Exercise 1.

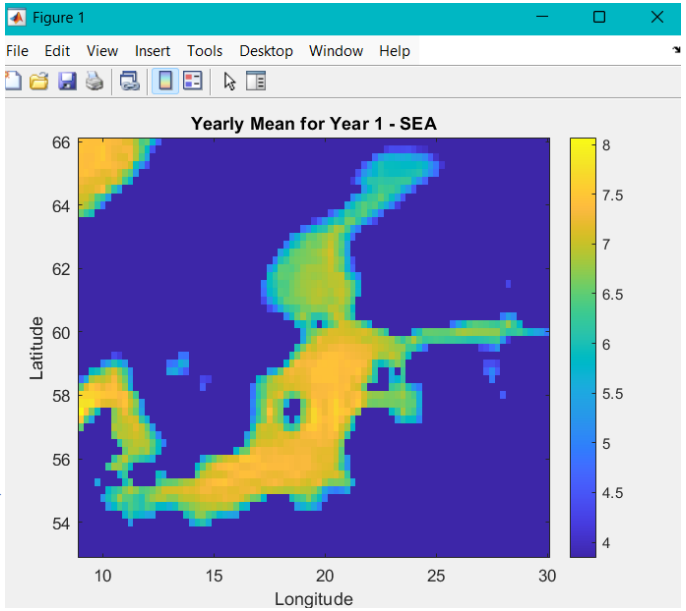
data_stream-oper_stepType-instant.nc

- ncread: time, lat, lon, u10, v10, sst, lsm
- Make a date vector: datetime
- Create grid of coordinates: meshgrid
- Calculate wind speed and direction (mathematical, North = 0 degrees, 360 degrees).
- Get yearly mean of wind speed for each grid cell.
- Plot one year of means: imagesc
- Use data on land-sea mask to mark land data as NaN.

Save your workspace.

Add descriptions to your code and save your code.



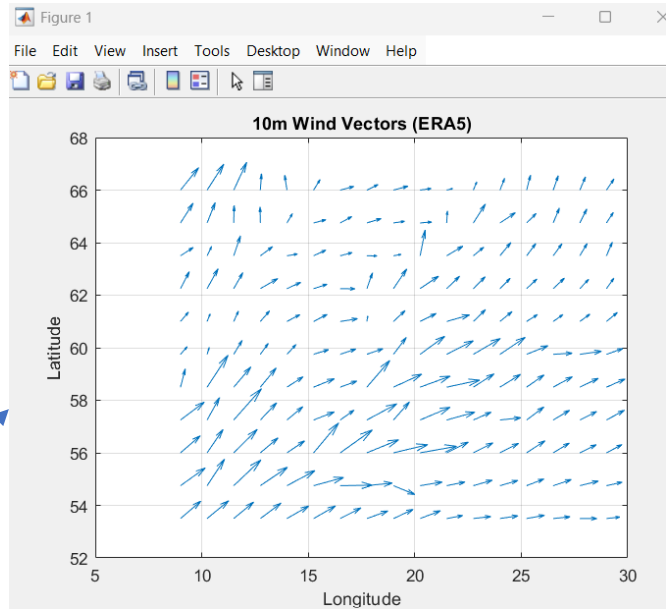


WS = sqrt($u10^2$+$v10^2$); WD = mod(atan2d(u10, v10), 360);

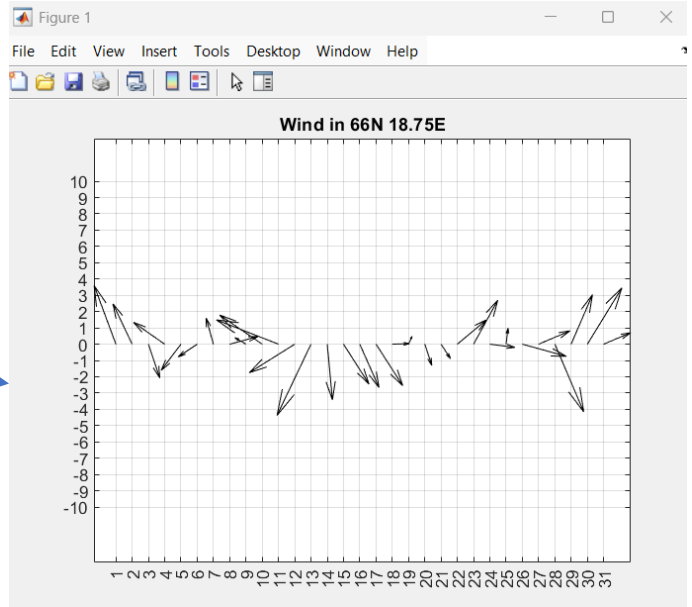
Lecture 6. Exercise 2.

Load the saved workspace from Lecture 6, Exercise 1.

- Get yearly mean wind components (u10, v10).
- Visualize u10 and v10 at every 5th coordinate using quiver.



- Get data from one month and one location (on sea), use quiver, axis equal



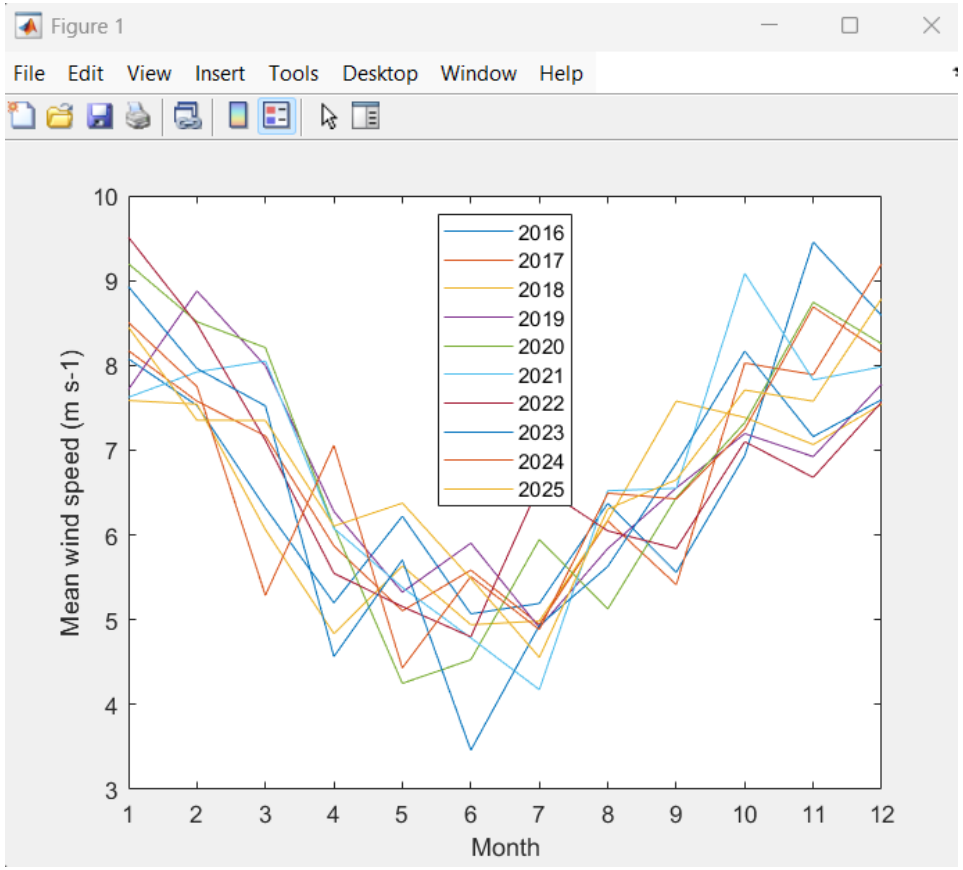
Add descriptions to your code and save your code.

Lecture 6. Exercise 3.

Load the saved workspace from Lecture 6, Exercise 1.

- Get monthly mean wind speed for every year from one location and plot it (y = wind speed, x = month)

Add descriptions to your code and save your code.

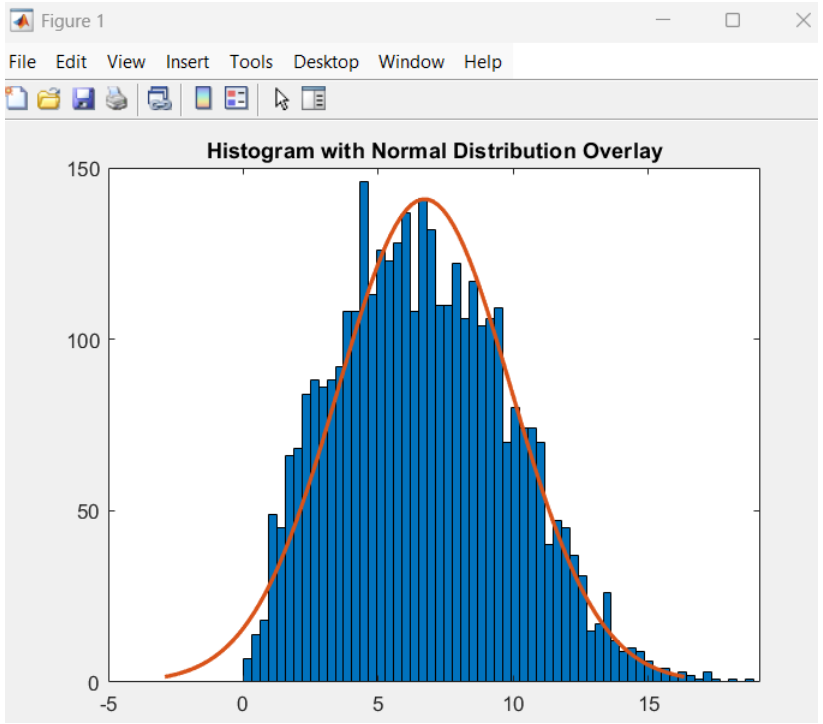


Lecture 6. Exercise 4.

Load the saved workspace from Lecture 6, Exercise 1.

- Get wind speed from one location and do a histogram with Normal Distribution Overlay:

  histfit



Add descriptions to your code and save your code.

Lecture 6. Exercise 5.

Load the saved workspace from Lecture 6, Exercise 1.

Get wind speed from one location.

% Fit the distribution

```
params = wblfit(wind_data);
scale = params(1);
shape = params(2); % Slope of the line
```

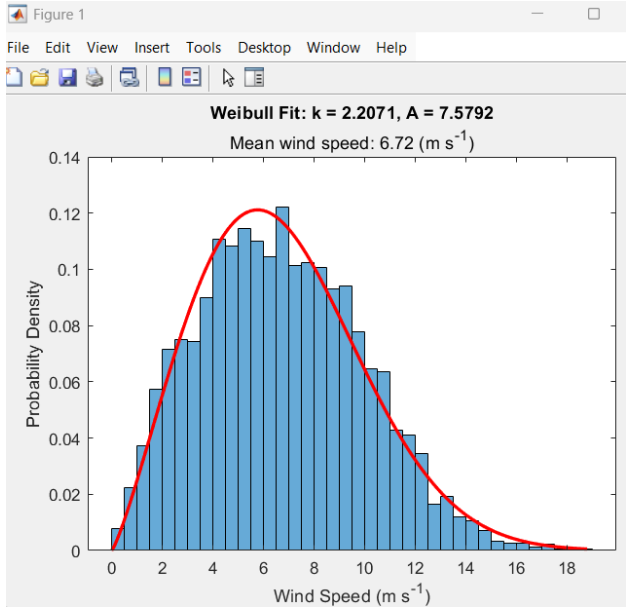
% Create a histogram

```
h = histogram(wind_data, 'Normalization', 'pdf');
hold on;
```

% Generate Weibull PDF values for plotting

```
x_range = 0:0.1:max(wind_data);
y_weibull = wblpdf(x_range, scale, shape);
```

% Plot the curve



The Scale parameter A tells you where cumulative probability reaches about **63.2%**.

If k < 2: You have a very wide range of wind speeds (highly variable).
If k ~3.5: The distribution actually starts to look like a Normal distribution.
If k > 3.5: The wind speed is extremely consistent (common in steady trade wind regions).

If Scale (A) is...	And Shape (k) is...	The Wind looks like...
High	High (>3)	Constant, powerful, steady winds (Trade winds).
High	Low (<1.5)	Frequent calms interrupted by massive, violent storms.
Low	High	Constant, light breezes (A quiet lake).
Low	Low	Mostly dead calm with occasional light puffs.

Lecture 6. Exercise 5. (continued)

Use your previously calculated parameters to calculate probability of a storm

```
A = scale;
k = shape;

% Threshold for a 'Storm' (e.g., 25 m/s)
threshold = 25;

% Probability that wind is <= 25 m/s
prob_below = wblcdf(threshold, A, k);

% Probability that wind is > 25 m/s (The 'Storm' risk)
prob_storm = 1 - prob_below;

fprintf('The probability of wind exceeding %d m/s is: %.4f%%\n', threshold, prob_storm * 100);
```

Add descriptions to your code and save it.

Lecture 6. Exercise 6

Load the saved workspace from Lecture 6, Exercise 1.

% Meteorological – wind from

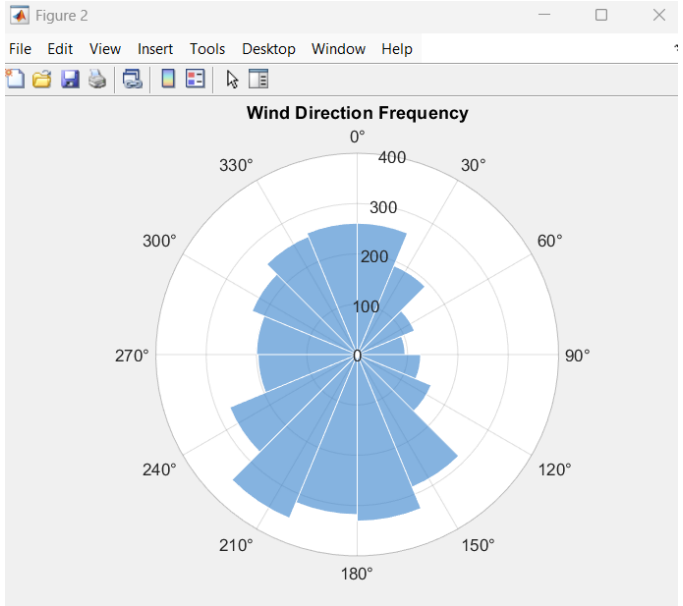
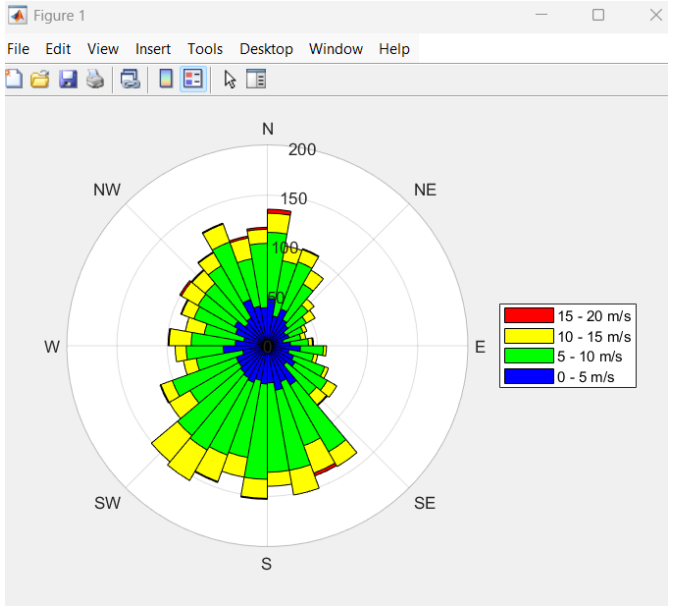
```
WD_from = mod(atan2d(u10, v10) + 180, 360);
```

% Mathematical - wind to

```
WD_to = mod(atan2d(u10, v10), 360);
```

- Create a wind rose

  Convert meteorological degrees back to radians for polar plot: deg2rad(WD_from)

  polarhistogram

Add descriptions to your code and save it.